Scientific Writing

Advanced Topics of Visualization and Communication Methods

Part 2

The structure of a paper

Let's discover a paper

- You must have read many papers but have you taken the time to analyse the way they are written, organized, their purpose, their intended audience
- Unlike a novel, a scientific paper MUST follow set rules about its structure and its content
- This is true too of a thesis or a scientific report
- Let's first analyze a paper by simply noting down the title of its various parts/sections…
- Bernard et al, 2016. Evidence for Eocene-Oligocene glaciation in the landscape of the East Greenland margin. Geology, v.44, pp.895-898.

Structure - Exercise 2

In the paper:

- In yellow highlight RESULTS
- In green highlight GENERAL STATEMENTS in Introduction
- Underline all parts dealing with METHOD description
- In red highlight REFERENCES cited in the discussion and introduction
- Highlights the different parts of the abstract (same color scheme)
- How would you qualify the last statement/paragraph of the paper?

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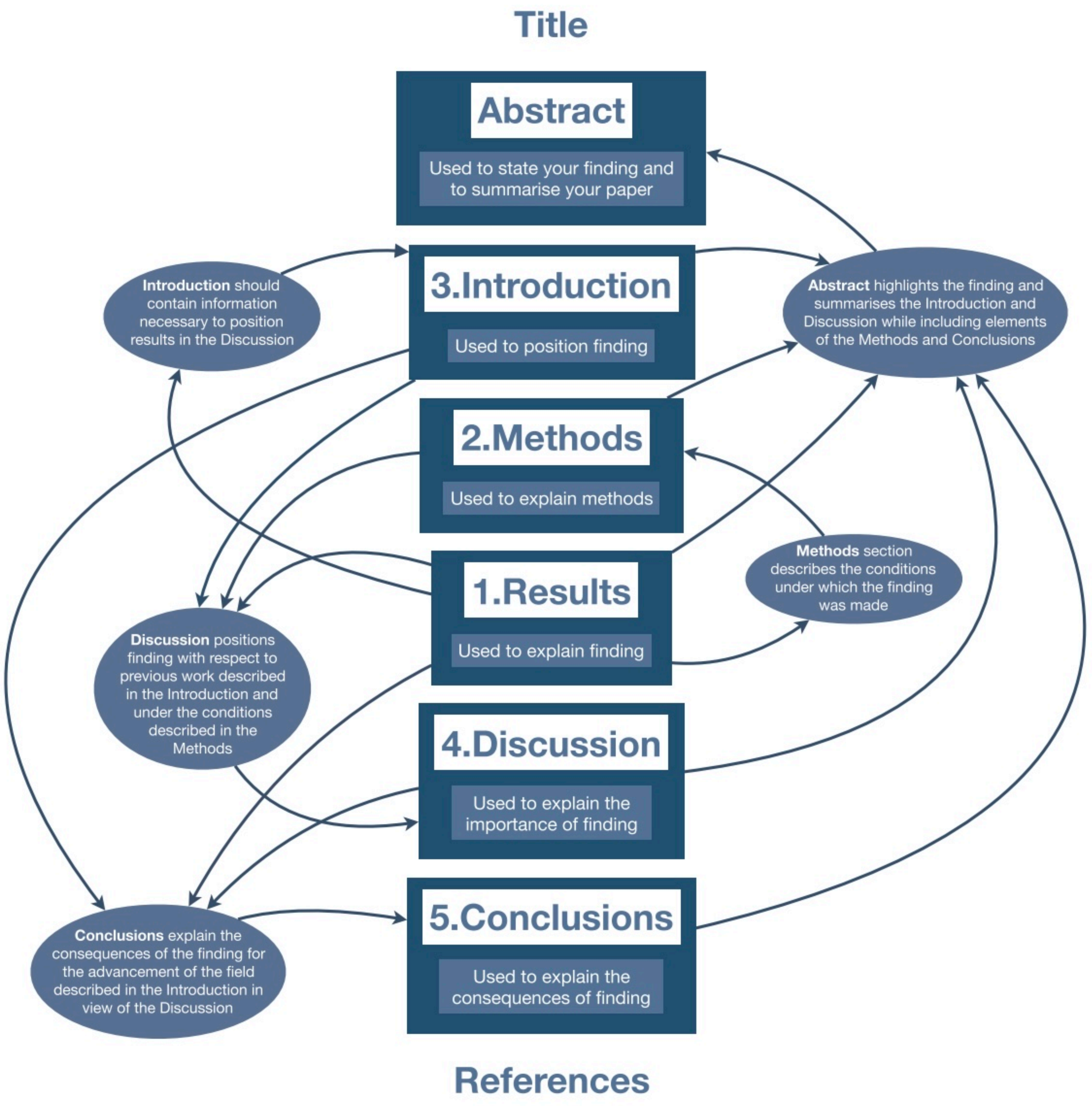
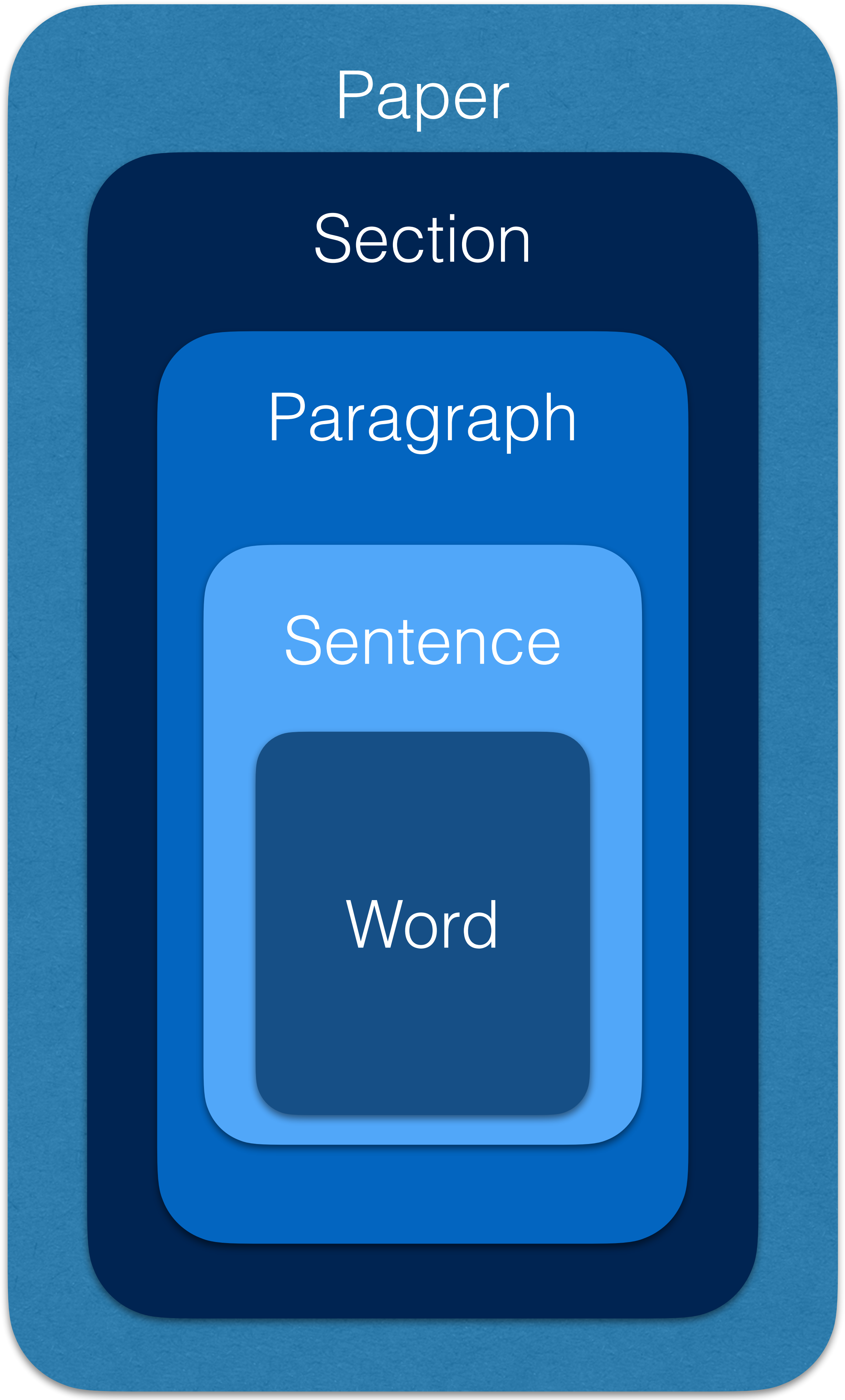


A paper is a collection of elements

- A section is the introduction, the methods, the results, the discussion, etc.; it must be composed of several paragraphs
- A paragraph is a logical unit which contain a single idea/concept; it must be composed of several sentences; it must contain a beginning, a body and an end.
- A sentence is a step in a logical unit; it must contain one verb, one subject and one object. Avoid long sentences.
- Words must be chosen for their clarity and precision; do not use spoken language; you are writing a scientific report, not a detective novel; do not hesitate to repeat a word.



Part 3

The introduction

INTRODUCTION

- RELEVANCE: a general statement concerning the field of investigation and indicating why it is relevant to the scientific community and/or larger community; you must attract the reader and/or getting him/her excited
- PREVIOUS WORK: the state-of-the-art, including a description of previous work, which might include different approaches and/or data; you must also demonstrate your knowledge of the field and your mastering of concepts, ideas, debates, etc. while limiting yourself to what is needed for the comprehension of the results and discussion
- NEXT STEP: what are the outstanding questions, or missing data/constraints? This prepares the reader to appreciate what you have done.
- BRIEF SUMMARY: what we intend to do

INTRODUCTION - Exercise 1

Introduction

TBW

Methods

To measure the ellipticity of the Earth, we fabricated two very long ropes of just above 20,000 km each. We positioned one of the two ropes from the North Pole to the South Pole and the second one around the equator. We adjusted the lengths of the ropes such that the first one measured the perimeter of the Earth around the equator and the second one along one of its meridians. We brought the ropes back to our laboratory where we carefully measured their lengths, which we called Δe and Δp.

We then measured the equatorial radius of the Earth, R_e, by using the relationship between the circumference and the radius of a circle: $R_e = \Delta e / 2\pi$. We used the following formula to obtain the polar radius of the Earth: $R_p = \sqrt{2\Delta p^2/\pi^2 - R_e^2}$.

Results

The results are R_e = 6378 km and R_p = 6356 km. From these values, we calculated a flattening $f = (R_e - R_p)/R_e$ = 0.0034.

Part 4

The Methods Section

METHODS

- In theory, you must provide all the necessary information for another researcher to reproduce what you are presenting in the paper
- Use previously published work to simplify this process, but provide basic information about methods for ease of understanding of the results
- Clearly describe any new methodological or technological development you have made
- Clearly describe assumptions, hypotheses and the limitations they might imply
- The value of all settings, parameters, constants, etc. must be provided (use of a table is recommended)
- Use appendices for complex or detailed methods

Methods - Exercise 2

Introduction

...

Our objective is to measure the change in tree diameter over one seasonal cycle. For this we developed a new tool, which, contrary to pre-existing ones, can be taken in the field.

Methods

TBW

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Part 5

The Results Section

RESULTS

- Built around a set of Figures and Tables (if necessary) and use ample reference to the figures in the text
- In constructing the figures, focus on displaying data/results in a meaningful manner such that it generates knowledge
- Each figure and item in a figure should have a clear purpose (a figure is not a stamp collection display)
- To achieve this, make use of data analysis software (correlation, spectral analysis, statistical analysis, etc..) or visualisation software (color contours, 3D plots of model results)
- Only show relevant data and results
- Focus on the basic results without an interpretation (commenting, not explaining)

Results - Exercise 3

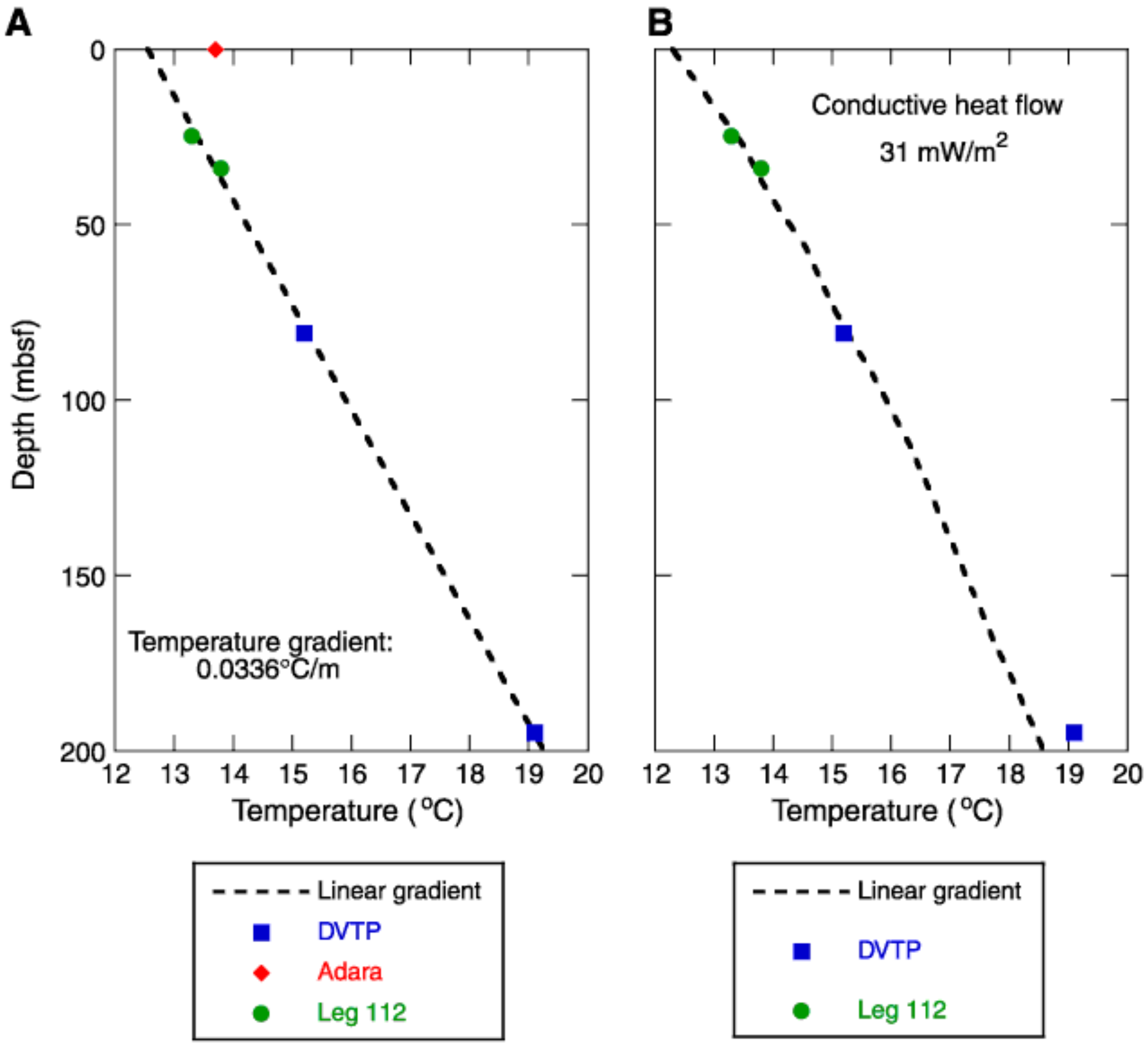

Figure F21. A. All temperatures measured in Hole 1228A together with the temperature data from Leg 112 Site 680 plotted vs. depth with a best-fit linear profile for the combined data. B. Temperature data with steady-state conductive temperature profile calculated using measured thermal conductivities from Hole 1228A and uniform heat flow.

Main rules for figures

- One idea per figure/panel
- Clarity (simple, simple simple,…)
- Efficiency (the main point must jump at the reader)
- Legibility (font, font size, color, green shades, line thickness, etc.)
- Legend (informative but not repetitive of the text; must help the reader to understand the figure, but it should not repeat the result)
- Figures must all be cited in the text and in the proper order (first citation decides of the number of a figure)

Tables

- Data tables can be long and complex
- They must sometimes follow a preset format
- Model/experimental parameters are usually given in a table
- If you perform experiments, use a table to summarize the different experiments and highlighting in what they compare/differ

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Part 6

The Discussion and Conclusions

Discussion

- Restate the original hypothesis/idea to be tested (go back to the introduction)
- Do your results support/disagree with it? In part, totally, to what degree?
- Why is it so? Explanations for the agreement/disagreement using other published work to support your explanation
- Remind the reader about the limitations of your study and why they may explain agreement/disagreement

Conclusions

- Implications for the field; be ambitious
- Recommendations for future work; be realist
- Include a “take-home” figure

Discussion & Conclusions - Exercise 4

Introduction

The Earth is almost round. Most believe that it is shorter around its spin axis. This has clear implications for how close one can fly planes around the Earth. But how flattened it is remains to be quantified. The first to measure this flattening was Aristotle (756 bc) who measures the distance to the horizons. From this he concluded that the Earth is nearly a sphere. Later, Lavoisier (1784) demonstrated that there is a measurable difference between the length of a meridian and a parallel. His estimate was however uncertain.In this paper we will improve on this result by using a new method we have developed. It consists in building two long ropes to directly measure the Earth's radii. Our results will be compared to existing measurements and used to determine how close planes can fly to the ground.

Methods

To measure the ellipticity of the Earth, we fabricated two very long ropes of just above 20,000 km each. We positioned one of the two ropes from the North Pole to the South Pole and the second one around the equator. We adjusted the lengths of the ropes such that the first one measured the perimeter of the Earth around the equator and the second one along one of its meridians. We brought the ropes back to our laboratory where we carefully measured their lengths, which we called Δe and Δp.

We then measured the equatorial radius of the Earth, R_e, by using the relationship between the circumference and the radius of a circle: $R_e = \Delta e / 2\pi$. We used the following formula to obtain the polar radius of the Earth:

$$R_p = \sqrt{2\Delta p^2 / \pi^2 - R_e^2}.$$

Results

The results are R_e = 6378 km and R_p = 6356 km. From these values, we calculated a flattening $f = (R_e - R_p)/R_e$ = 0.0034.

Discussion:

TBW

Conclusions:

TBW

Part 7

The Abstract

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ABSTRACT

- Start with a catchy or informative sentence that highlights the relevance of the work
- Must be an accurate summary of the paper: relevance of the work, methods used and/or main hypotheses, results and conclusions (implications, limitations)
- It is commonly accepted to spend one half of the abstract on the results/conclusions
- The abstract should contain a sentence starting with: “We demonstrate that” or “We show that” or “We introduce a new notion/parameter/idea”

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Abstract - Exercise 6

Introduction

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Results

The results are R_e = 6378 km and R_p = 6356 km. From these values, we calculated a flattening $f = (R_e - R_p)/R_e$ = 0.0034.

Discussion:

Using our new method, we have obtained a measure of the ellipticity of the Earth that is more accurate than any previous measurements (Aristotle, 756 BC, Lavoisier, 1784). Our estimates is three orders of magnitude more precise, because we used a rope the length of which can be directly measured in the lab.

However, we realise that our measurements were affected by the topography of the Earth along the meridian that we selected. For example, if the meridian had crossed the Himalayas, we would have obtained a different measurement. Furthermore, the Earth is know to change its shape with the time of the year.

This implies that future measurement campaigns are required along different meridians and at different times of the year. This would certainly decrease the uncertainty of estimate of Earth's flattening.

Conclusions:

We have made an accurate measurement of Earth's flattening and it is 0.0034.

This implies that we can now fly planes 200 m lower than previously accepted. However the is an average figure that needs to be perfected locally by further measurements.

Part 8

Other parts

TITLE and AUTHORSHIP

- The title should be informative and attractive
- Authorship should be a reflexion of the work done, including conceptualizing of ideas/concepts, performing the measurements and/or field work, developing methodology (including writing software), synthesizing the data and describing the results, preparing the manuscript (writing the text, preparing the figures and tables)
- Many journals now require that the contribution of each author be properly described

Title - Exercise 7

At the end of the Miocene, the European Alps ceased outward expansion, and tectonic uplift and exhumation shifted into the orogen interior. This shift is consistent with a change from orogenic construction to orogenic destruction, reflecting an increase in the ratio of erosional flux to accretionary flux. The coincidence of this change with an increase in sediment yield from the Alps suggests a climate-driven increase in erosional flux. The timing of deformation and sediment release from the southern Alps indicates that the tectonic change occurred synchronous with the last phase of the Messinian salinity crisis. We attribute the increase in erosional flux to a climatic shift to wetter conditions throughout Europe, likely augmented by the base-level fall that occurred during the Mediterranean dessication. This climate change is represented in the stratigraphic record by the Lago Mare deposits of the Mediterranean salinity crisis.

Solution

- Messinian climate change and erosional destruction of the central European Alps

OTHER BITS

- Acknowledgements: include the name of those who have made a small contribution (by proof-reading the paper, the reviewers, etc.), cite funding sources
- References (see below)
- Figure and Table captions (see below)
- Appendices: any information that might be necessary to comprehend the results but would “break the flow” of the paper
- Supplementary material: data tables, details of equipment used, in some journals any material that, due to their length, could distract from the main “logic” of the paper

Part 9

Workflow

Writing workflow

- Where do I start?
- In what order do I write the different sections?
- How long will it take to prepare the paper?
- **Exercise: write a short calendar for writing the different sections of the paper**

Suggested workflow

- Main figure(s) reporting main finding(s)
- Results section
- Methods section
- Introduction section
- Discussion section
- Other bits
- Abstract

Write your paper

1. write down a short paragraph summarizing the main result(s) of your study
2. draft a figure explaining/reporting this result
3. decide which journal you are going to submit this paper to and check author's guidelines for this journal
4. draft a plan of your manuscript (using or not the main template discussed above)
5. decide where the figures will go and what they will look like
6. decide on a writing plan/calendar
7. start writing

Random bits

Citing

- Two things are important about references: where/how to cite and how to write the citation.

- A citation’s purpose if to inform the reader about previously published work, give proper credit to the author of a work, an idea, a data an observation or a theory. It is therefore critical to include the citation immediately after the word/sentence that describe the work, idea, data, observation or theory.

- If several authors have worked on the same idea or collected similar data, one must refer to all relevant work and give them in the appropriate order (usually the chronological order). In practice, you separate the different citations by a semi-colon.

- It is sometimes convenient to cite a review article, especially in the introduction, to define the state-of-knowledge, but it cannot be used without citing the most important works first.

Citing

- You can use Braun (2012) or (Braun, 2012). The former implies that the citation is an active element of the sentence (subject, object); the later is used when the citation qualifies a word, idea, sentence.

- Examples:

  - Braun (2012) showed that the Earth is essentially a sphere. According to Braun (2012), the Earth is flat.

- But

  - The Earth is a sphere (Braun, 2012).

Citing

- Referencing differs from journal to journal but in general it also varies with the number of authors:
  - I. one author : (Braun, 2000);
  - II. two authors : (Braun and Bormans, 2000);
  - III. more than two authors : (Braun et al., 2000).

Examples of good/poor citing

Poor citing

- Recent studies have demonstrated that whales were larger than dolphins but less grey than mice (Braun, 2010; Braun, 2012)

Good citing

- Recent studies have demonstrated that whales were larger than dolphins (Braun, 2010) but less grey than mice (Braun, 2012)

or

- Recent studies (Braun 2010; 2012) have demonstrated that whales were larger than dolphins, but less grey than mice.

References

- Varies from journal to journal
- Always use the appropriate format (format file usually provided by the journal)
- Many different types of references exist and each has its own format (journal, book, chapter in book, PhD thesis, map, data repository, etc.)

On the use of the passive form

- What should we use?
  - We have used the method by Braun (2012).
  - The method described in Braun (2012) was used.
- Ask yourself the question: is it important to know who performed the action (in this case “use”). If the answer is “yes” the passive sentence requires the use of “by …” and should therefore be avoided

Which tense to use?

- Be consistent in a sentence, paragraph, section, paper
- The present should be used to describe parts of the paper (a figure, a table, etc.) as well as widely accepted ideas (truths)
- The past must be used to described specific previously published work or steps taken during the study

A few rules

- Avoid the use of unclear pronouns ; example of poor use of a pronoun
  - “Mammal habitats are very dispersed, contrary to bird habitats, except for those which are close to the equator.”
- You cannot start a sentence with a number
- A list (or bullet point) must be homogeneous and contain the same type of elements. Example of a poor list:
  - “The procedure included:
    - sample crushing,
    - mineral separation,
    - to measure U concentration,
    - we then measure He concentration to determine ages.”